

DNYANESHWAR SAHANE



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🌐 [Dnyaneshwar Sahane | LinkedIn](#)

OBJECTIVE

To secure a leadership position in the emerging electronics sector where I can leverage my extensive technical expertise, innovative mindset, people management and proven experience in managing complex projects. I aim to drive the development of cutting-edge technologies, foster collaborative team environments, and contribute to meet the strategic goals of the organization while continuously enhancing my skills and knowledge in the rapidly evolving technological landscape.

PROFILE SYNOPSIS

A competent and professional Engineering Leader with over 16+ years of experience, including more than 5+ years in managing and leading diverse, cross-functional teams in electronics product design and development. I bring extensive expertise across domains such as power electronics, welding, motor control, packaging, security systems, defense, electric vehicle chargers, energy management, commercial aerospace electric actuation, healthcare, and industrial products. My core strengths lie in electronics hardware and firmware development, including FreeRTOS and interrupt-based firmware, bare-metal coding, and device driver development. I have hands-on experience in implementing communication protocols, designing software-hardware architectures, and developing bootloaders, CPU stack, heap, and memory management solutions. My technical proficiency includes bootstrap, U-Boot, embedded Linux porting, and device tree source configuration, along with Linux device drivers, network drivers, kernel code customization, shell scripting, make files, and root file system development.

I am well-versed in technologies such as LILO, IoT, MODBUS TCP, Node-Red, and have developed algorithms for BLDC motor control, crane control, packaging, welding, stepper motors, power electronics, telecom, fencing, eHASP, canister mechanisms, secure computing, vehicle tracking, and access control systems. My capabilities span both analog and digital circuit design, embedded platform debugging, and compliance with electronics design standards. As a people-centric leader, I excel in building, mentoring, and managing high-performing engineering teams. I foster a collaborative and accountable work culture, drive team engagement, and align individual strengths with organizational goals. I have successfully led geographically distributed teams, facilitated knowledge sharing, and implemented structured performance evaluation and career development frameworks.

In my leadership role, I am adept at coordinating cross-functional teams, managing the software development lifecycle (SDLC), and implementing Agile Scrum methodologies. I am committed to tracking project progress, resolving technical challenges, and ensuring adherence to business process standards such as CMMI Level-3 and RoHS. My experience also includes preparing technical proposals and bids, collaborating with vendors and international technical partners, and maintaining strong customer relationships. I am proficient in creating comprehensive technical documentation, including detailed technical proposals.

TECHNICAL FORTE

Programming Languages: Assembly & Embedded C/C++, Python, Scripting, Makefile.

Microcontrollers/Processors: 8/16/32 bits

- ✚ Atmel AVR Mega series- AT90CAN128 | ATMEGA8, AVR Mega | Xmega series
- ✚ Microchip-PIC16, PIC32, dsPIC, SiLab-MCS-51, Renesas: 32 bits controller.
- ✚ TI-C2000, Single core-TMS320F28233, TMS320F28335 and Dual core-TMS320F28779D, F28M35H52C
- ✚ NXP-LPC1768/88, LPC4357, LPC4088, LPC2368, LPC2148, P89V51RD2, P89LV51RD2
- ✚ STM- STM32F103, STM32F412/13, STM32F091RC, STM32F030R8. TI: Stellaris Cortex M3 LM3S5R36
- ✚ Intel: Quark x1000, TI: AM3352, ATMEL: SAMA5D27

Basic knowledge of FPGA and programming by Xilinx product SPARTAN-6

Operating System: Windows, FreeRTOS, Basic knowledge of Linux.

IDE | Compilers: KEIL IDE, Code Composer Studio, Control suite, Atollic TRUEStudio, MPLAB X, Atmel Studio, Code Blocks, Visual Studio, Visual Studio code, eclipse, DevC++, Simplicity Studio, STM32CubeIDE, STM32CubeMX.

Design Tools | Simulation: Orcad, Altium, KiCad, Pspice, Pads, Circuit Maker, TINA (TI), Proteus, LT Spice.

Product Life cycle Management: Wrench PLM, Team Center.

Documentation Tools: Doxygen, MPP, SVN, Clear case, GitHub, Git, Gitlab, JIRA, Nexus.

Communication Buses: Ethernet, I2C, SPI, RS232, RS485, RS422, CAN, FSMC, Parallel port, SSI.

Communication Protocols: TCP|UDP, MODBUS, AT Commands: GSM | Bluetooth, CANOpen, CTA2045.

Electronic Modules/Devices: Stepper | DC | PMSM motor drivers, GPS, GSM, Bluetooth, FRAM, SRAM/NVRAM, SPIFI Flash, E2PROM, FLASH, SD Card, Bluetooth, A/D Encoders, USB keyboard, ADC, DAC, Resolver, Tachometer, Expanders, Latches, Buffers, Displays, Light | Temperature | Proximity | Current | Liquid | Pressure | Air | IR | Speed | Ultrasonic sensors, Liquid | Air | Gas pressure valves | Wifi.

Equipment's: DSO/CRO, Scope coder, Logic Analyzer, Power Analyzer, Function Generator, Electronics load, Digital Multimeter, Thermometer, CAN Analyzer, Multi-function power supply source/load.

JTAG Debugger: U-Link, J-Link, U-Link-Pro/D, ST-Link, GDB, XDS100/200, USB560/2000, Pickit3/4, AVR JTAGICE.



FUNCTIONAL SKILLS

- | | |
|---|---|
| ✚ Customer & Stakeholder Management | ✚ Capex and OpEx Management |
| ✚ Technical Leadership & coaching | ✚ Compliance and Standards |
| ✚ Goals setting and expectation | ✚ Technical Documentation |
| ✚ Team & Performance Management | ✚ Embedded HW & SW Development |
| ✚ Team Hiring & Talent Career Development | ✚ Cortex-ARM Microcontrollers, FreeRTOS |
| ✚ Cross Functional collaboration | ✚ AVR & PIC Microcontrollers |
| ✚ Planning, resource allocation, budget, timeline | ✚ 8/16/32 Bit Microcontrollers |
| ✚ Product development & assurance | ✚ Digital & Analog Electronics |
| ✚ Innovation and Analysis | ✚ Embedded Linux development |
| ✚ NPD, NPI, NTI Product Development | ✚ Embedded C/C++ Programming |
| ✚ Project Management | ✚ MODBUS, UART (RS232 / RS422 / RS485) |



ELECTRONICS STANDARDS

IEC62304, ISO14971, ISO13485, ISO9001:2008, IEC61851, VDE-AR-N 4105, IEC60664-1, ISO15118-1-3, ISO26262, ISO11898-1: 2003, MIL-STD-461C-F, MIL-STD-498, CTA2045, DO160, DO254, DO178.



PROFESSIONAL EXPERIENCE



Engineering Leader | Eaton India Innovation Center, Pune, Duration: since April 2023- till now.

Key Responsibilities:

Leadership & Team Management

- ✚ Leading a team of 20+ engineers in a Center of Excellence for aerospace, electrical selector, and mobility product development.
- ✚ Responsible for team profit planning, performance management, talent development planning, competency building, training, mentoring, and coaching.
- ✚ Foster a collaborative, innovative, and high-performance work environment.
- ✚ Lead and mentor engineering teams in the design and development of electronic products.
- ✚ Serve as the Single Point of Contact (SPOC) for global teams and stakeholders.

Program & Project Management

- ✚ Leading customer-funded programs across aerospace clients, including IRAD and ITAR-compliant projects.
- ✚ Managing multiple concurrent projects including NTI (New Technology Introduction), NPI (New Product Introduction), and NPD (New Product Development).
- ✚ Currently overseeing four NTI programs focused on electric actuation systems for business jets and commercial aircraft.
- ✚ Manage project timelines, budgets, resources, and deliverables to ensure on-time and within-budget execution.
- ✚ Conduct risk assessments and implement mitigation strategies.

Cross-Functional Collaboration

- ✚ Work closely with global business teams, local cross-functional teams, suppliers, and supply chain management (SCM).

- ✚ Collaborate with system simulation, motor design, mechanical, software, SCM, and project management teams.
- ✚ Maintain regular communication with customers and stakeholders, providing feedback on deliverables and team performance.
- ✚ Coordinate with marketing, manufacturing, and quality assurance teams to align project goals.

Product Development & Technical Oversight

- ✚ Oversee the entire product development lifecycle—from concept to production.
- ✚ Ensure designs meet technical specifications, safety standards, and regulatory requirements.
- ✚ Develop and validate prototypes; refine designs based on testing outcomes.
- ✚ Manage lab infrastructure and operations to support development and testing activities.

Process & Quality Management

- ✚ Establish and benchmark standard processes and engineering guidelines.
- ✚ Implement and maintain quality assurance processes to ensure high standards.
- ✚ Ensure compliance with industry regulations and standards (e.g., aerospace, safety, and quality norms).

Innovation & Continuous Improvement

- ✚ Drive innovation by exploring new technologies and methodologies.
- ✚ Continuously improve design processes, tools, and team capabilities.
- ✚ Promote a culture of continuous learning and improvement within the team.

Stakeholder & Vendor Engagement

- ✚ Communicate project status, challenges, and solutions to internal and external stakeholders.
- ✚ Collaborate with clients to understand requirements and ensure satisfaction.
- ✚ Interact with vendors and partners to ensure smooth collaboration and project execution.

Technical Responsibilities:

- ✚ Lead the preparation and presentation of business proposals to support strategic initiatives.
- ✚ Conduct in-depth analyses and trade studies to assess and select new technologies for implementation.
- ✚ Drive, participate in, and approve technical design reviews, product proposals, and test result evaluations.
- ✚ Directed the development of hardware, software, verification & validation (V&V), and systems engineering teams for projects in the aerospace and electrical sectors.
- ✚ Spearheaded algorithm development efforts to enhance product functionality and performance.
- ✚ Managed all phases of the product development lifecycle, including requirements gathering, design analysis, implementation, prototyping, testing, verification, and validation.

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PROJECTS

eTRAS, Spoiler, HePP, PW100, HEW50, PDU, NAC, TMS, MIC, SMPS, Pisgah, AILE, FS2.



Software Designer – Embedded System | Philips Healthcare LTD, CT/AMI R&D, Pune

Duration: since Feb 2022-Mar 2023.

Key Role:

- ✚ Developed firmware for the advanced motor controller to operate within legacy systems using VxWorks RTOS.
- ✚ Successfully developed and integrated a common motor controller for all CT/AMI machine applications.
- ✚ Made significant contributions to the redevelopment of the patient table motor controller, enabling legacy system integration with an advanced controller for unified use.
- ✚ The development and enhancement of CT and AMI product modalities, with deep system-level understanding of CT systems.
- ✚ Identified and resolved customer issues through innovative solutions, providing continuous support and guidance to the technical team.
- ✚ Oversaw software and hardware integration, including system software platform development, technical design, and code reviews.
- ✚ Championed “Customer First” initiatives and ensured compliance with relevant medical standards and regulations.
- ✚ Managed legacy obsolescence replacement and evaluated feasibility for new designs, contributing to the development roadmap for CT and AMI products.
- ✚ Played a key role in team engagement, training, mentoring, and planning of technical activities.

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PROJECTS

- ✚ CT/AMI – SPECT System: Brightview X



R&D Manager | Siemens LTD, Global R&D, Mumbai, Duration: since Oct 2019 – Jan 2022.

Key Role:

- ✚ New product requirements, feasibility analysis, cost analysis, firmware application, logic algorithms development, HW | SW architecture, communication protocols, system architecture, flowcharts, DFMEA, Boot loader, system testing & integration, power electronic control algorithms.
- ✚ Capturing global project requirements, team management, technical proposals (project IDC), technical documentation, project estimation, project planning, project schedule, project execution, risk analysis, product security, business process and guideline, weekly project status and meeting with Germany team.
- ✚ Frequent interaction with global R&D team, Compliance team, business team, PLM team, Senior management, Acting as SPOC to the customer for all technical and business-related issues.
- ✚ Technical support to the team for Firmware & hardware design and system interface. Technical review, organize resources, task allocation, guidance to team, daily resource task status update.



Projects

Electric Vehicle Charger Controller

- ✚ Responsibility: Requirements, HW/SW Architecture, Electric vehicle charger controller circuit design which includes AC-DC & DC-DC converter, charger panel, HMI, PLC interface, algorithm development, application firmware, control board design, testing, integration, and project execution.

Power Monitoring Unit

- ✚ The product is used to control the 3- phase grid voltage, power, frequency monitoring and PV inverter control according to country standards. Responsibility: Start to end of product solution.

VersiCharge: Electric Vehicle AC charger, wall mounted.

- ✚ Firmware development for the Microcontroller Cortex-M0 and Linux processor. Communication between Microcontroller-M0 and Cortex-A8 over SPI communication using CTA2045 standards. EV charging standards - CCS type 1 and type 2 AC/DC, CHAdeMo.



Technical Manager- Embedded Design | L&T Technology Services LTD, Mumbai

Duration: Nov 2017 - Oct 2019

Key Role:

- ✚ Direct client interaction & meetings, Project status report, Client requirements, deliverables tracking.
- ✚ Managed a team size of 8 to 12 for project execution like hardware, firmware, integration & testing.
- ✚ Replicated the project specific requirement specification for electronics HW-SW, design calculation, value engineering.
- ✚ Technical proposals, cost estimation, budget, schedule, team management, SPOC for customer.
- ✚ Responsible to development firmware, Algorithms, SW/HW Architecture, flowchart, system communication protocols, coding standards, code, communication interface testing & integration.
- ✚ Technical support to team, technical reviews, problem analysis and solution, suggestions.
- ✚ Developed Algorithms: Welding, Wire feeder, Protection, Liquid control, Film Control, Packaging.
- ✚ Technical documentation: DOU, SOW, PSR, Proposal, Test Plan, Test Report, RS, HLDD, DLDD.



PROJECTS

Automated Packaging Machine

- ✚ Managed a team size of 5-8 for the design and development of the entire project.
- ✚ Design and development of hardware, application firmware, Algorithms for following modules.
 - Liquid Control, Bag Seal Control. Film Control, Vent Motor Control, Jaw Motor control.



Boot loader Development

- ✚ Development of Boot loader for Cortex M4 based devices (STM32F412/13).
- ✚ Encrypted firmware technique using GUI over serial RS232 interface.

Battery Operated Automatic Welding Machine using HF TiG

Robotic Welding power source which can weld the Gas pipelines from the internal side. This is the world's first automated robot which works on customized designed batteries with water-cool technology. 30V/210A power source for HF -TIG/MIG/Pulse welding. Customer patented this product.

- ✚ Design and development of Embedded HW, algorithm, and application firmware for the e-modules.



Sr. Embedded Design Engineer | The Tata Power Company Ltd, SED R&D center, Mumbai

Duration: Sep 2012 – Nov 2017 (5 Years, 2 months)

Key Role:

- ✚ Start to end in house embedded system product solutions for different defense projects.
- ✚ Developed Algorithms and Firmware using Embedded C & RTOS for different applications.
- ✚ Responsible for replicating the requirement specification in the system by design and development of electronics products with mil standards and compliance.
- ✚ Communication protocols, communication interface with external modules, design microcontroller-based circuits, analog & digital signal requirements.
- ✚ Software architecture, Motor control Algorithm, System algorithms/flowchart, Module test codes for individual PCB testing, development of firmware to system integration and testing.
- ✚ Production and testing documentation, Component's part creation in Team center (PLM), component selection, costing budget, Component selection, bill of material.
- ✚ System integration and testing with launchers and guns, wiring chart, cable harness.
- ✚ Design review by the senior technical team, detailed understanding of requirement, Demonstration and trial at military areas on-site, environmental testing, Vibration, Shock, functional testing, qualification, and validation tests & Technical documentation.



PROJECTS

155mm Advanced Towed Artillery Gun System (TAGS)

- ✚ The Advanced Towed Artillery Gun System (TAGS) is a towed 155 mm/52 caliber howitzer that is being developed for the Indian Army by DRDO in collaboration with private sector.
- ✚ Design and development of motor control Algorithms for running a PMSM motor in close Loop with Resolver & Tachometer feedback for different mechanism control applications.

Prahar Missile Launcher System

- ✚ Prahaar ("Strike") is an Indian solid-fuel road-mobile tactical ballistic missile developed by the DRDO. Prahaar is expected to replace the Prithvi-I short-range ballistic missile in Indian service.

Integrated Electronic Warfare System

- ✚ SRC's integrated electronic warfare (IEW) systems provide ground forces in mobile and fixed locations with the ability to seize, retain and exploit an advantage within the electromagnetic spectrum.

Telecom Power Management System

- ✚ This product is specific for the telecom infra industry ATC India. This product is deployed on the village site in different sites of India.

Secure Desktop Computer

- ✚ Development of firmware for secure boot microcontroller for Secure Desktop.
- ✚ Interfacing with i-Button for authentication and X-Wall for SATA encryption.

eHASP-iLOTO

- ✚ eHASP is the remote control automated authorized digital control unit with TCP and GSM replaced the physical iLOTO mechanical system.

Fence & Security System

- ✚ This Product is deployed on Tata Power SED, Mumbai location site and it is working perfectly.



Embedded Design Engineer (R&D) | Track RF Link Private Limited, Borivali

Duration: Jan' 2010 – Aug' 2012 & Aug' 2006-Aug'2007 (3 years, 2 months)

Key Role: Firmware development in Embedded C, module testing code, Schematic design, design calculation, debugging, defining testing procedure, manufacturing prototypes, feasibility study, Bill of material, PCB design, PCB verification, bench marking, DFMEA, Antenna Design for RFID modules, production and testing team support.



PROJECTS

- ✚ RFID based attendance monitoring system.
- ✚ RFID based toll reader.
- ✚ Bus ticket validation RFID smart card reader.

- ✚ Vehicle tracking system using GPS & GSM.
- ✚ Bluetooth based thermal printer.



PAPER PUBLISHED

- ✚ A paper on "Indoor farming using Solar Energy" to MD. Tata Power for Innovative ideas. The paper was selected for the first prize. The idea was implemented indoors and successfully verified.
- ✚ Design of a low-cost welding machine controller with a novel control algorithm for an enhanced HF TIG welding process. <https://iopscience.iop.org/article/10.1088/2631-8695/ac6fb5/meta>



AWARDS RECEIVED

- ✚ Received a country award in the Siemens LTD in 2021.
- ✚ Received a star employee award in the Siemens LTD in 2020.
- ✚ Awarded for Team Champion in the L&T Technology Services in 2018.
- ✚ Two times awarded for Outstanding Performance of the Employee in the Tata Power SED in 2015 & 2017.



CERTIFICATES

- ✚ DFMEA & SFMEA internal Philips training.
- ✚ Advance Embedded Linux development by Coursera
- ✚ Advance C programming by LinkedIn
- ✚ IoT Foundation Fundamentals by LinkedIn
- ✚ Problem Solving Techniques by LinkedIn
- ✚ AWS Fundamental Going Cloud Native by Coursera
- ✚ Software Development Processes & Methodologies by Coursera
- ✚ Pointer & Memory Management in C/C++ by Coursera
- ✚ Programming Fundamental by Coursera
- ✚ Introduction to Electric vehicle by Udemy
- ✚ Introduction to Linux and command line by Udemy
- ✚ Introduction to IIOT by NPTEL
- ✚ Introduction to Embedded System Design by NPTEL
- ✚ Lean Software Development by Coursera
- ✚ Agile Software Development by Coursera
- ✚ All PMP modules Course by Coursera



EDUCATION

- ✚ 2021 MTech in VLSI and Embedded System from the JSPMs Rajarshi Shahu COE, Pune. Obtained 9.31 CGPA.
- ✚ 2010 B-Tech in Electronics & Telecommunications from MIT, University of BAMU Aurangabad with 81%
- ✚ 2006 Diploma in Electronics & Communications from Amrutvahini Polytechnic MSBTE with 72.24%



PERSONAL INFORMATION

- ✚ Date of Birth: 6th April 1985
- ✚ Marital Status: Married.
- ✚ Nationality: Indian.
- ✚ Gender: Male
- ✚ Passport No: R9279199
- ✚ Language known: English, Hindi, Marathi



DECLARATION

I hereby Declare that all the above given information is true to the best of my knowledge and belief.

Place: Pune
Date: 2025

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